

Project Title

Full Stack E-Commerce Web Application

1. Project Overview

This project is a **Full Stack E-Commerce Web Application** developed using **React.js, Node.js, Express.js, and MongoDB**. The application provides an end-to-end online shopping experience including user authentication, product browsing, cart management, It demonstrates real-world implementation of scalable MERN stack architecture.

2. Objectives of the Project

- To design and develop a real-world e-commerce platform
 - To implement secure user authentication and authorization
 - To manage products, carts, orders, and users efficiently
 - To integrate payment gateways for online transactions
 - To apply industry-level project structure and best practices
-

3. Technology Stack

Frontend

- React.js
- React Router DOM
- Context API (Auth & Cart Management)
- Fetch API
- CSS

Backend

- Node.js
- Express.js
- RESTful APIs

Database

- MongoDB (NoSQL)
- Mongoose

4. Project Architecture

The project follows a **Client–Server Architecture**:

- Frontend (React) handles UI and user interactions
- Backend (Node + Express) handles APIs and business logic
- MongoDB stores application data (users, products, orders)

5. Functional Modules

5.1 User Module (Customer)

- User Registration & Login
- Email verification & password reset
- Profile management
- Wishlist management
- Cart management

5.2 Product Module

- Product listing with pagination
- Product details with images and pricing
- Search, filter, and sort functionality

5.3 Order & Checkout Module

- Address selection
- Payment methods (COD / Online Gateway)
- Order confirmation & tracking
- Invoice generation (PDF)

5.4 Admin Module

- Admin authentication
- Dashboard analytics
- Product management
- Order status management
- User management

6. Core Features

1. Secure user authentication
2. Product browsing with search & filters
3. Cart & wishlist functionality
4. Online and offline payment support
5. Admin dashboard for full control

7. Project Directory Structure

Backend Structure

```
ecommerce-app/  
├── backend/  
│   ├── src/  
│   │   ├── config/  
│   │   ├── models/  
│   │   ├── controllers/  
│   │   ├── routes/  
│   │   ├── middlewares/  
│   │   ├── utils/  
│   │   ├── app.js  
│   │   └── server.js  
│   ├── package.json  
│   └── .env
```

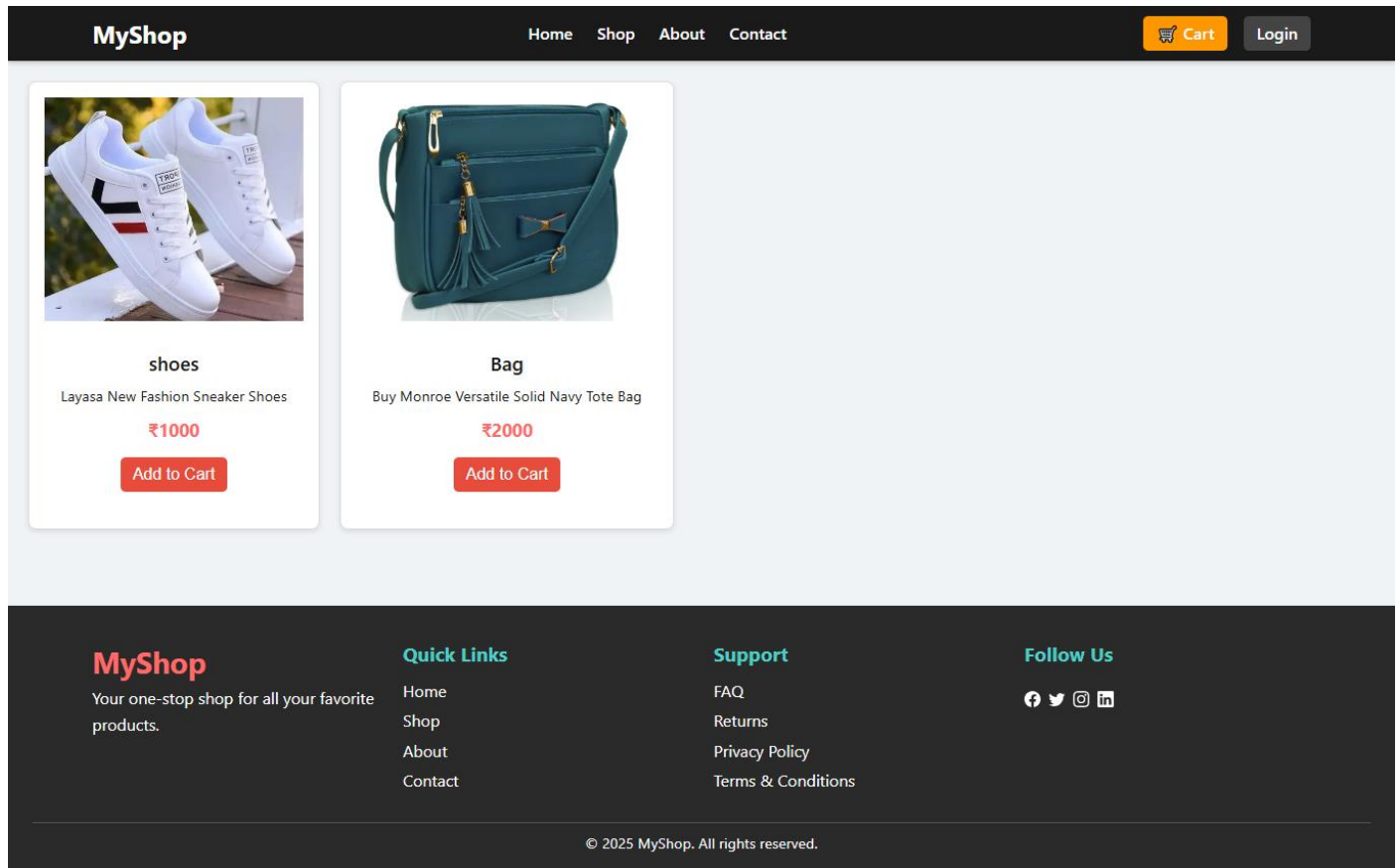
Frontend Structure

```
frontend/  
├── src/  
│   ├── assets/  
│   ├── components/  
│   ├── pages/  
│   ├── context/  
│   ├── hooks/  
│   ├── services/  
│   ├── utils/  
│   ├── App.jsx  
│   ├── main.jsx  
│   └── index.css  
├── package.json  
└── .env
```

8. User Interface Description

Home Page

- Product listings with categories and filters



Register Page

Register

Register

Already have an account? [Login](#)

Product Detail Page

- Product images, price, stock, and description

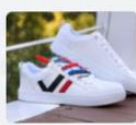
MyShop[Home](#)[Shop](#)[About](#)[Contact](#)[Cart](#)[Login](#)



Layasa New Fashion Sneaker Shoes
₹1000
In Stock
Step out in style with these trendy fashion Featuring laped to day anan design with and comfortale sole a comfortable sole. Perfect Available in multiple sizes.

Size: Quantity:

[Add to Cart](#)



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Cart Page

- Added products with quantity update option

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Cart Page



Layasa New Fashion Sneaker Shoes Shoes

Quantity: Total: ₹1000



Buy Monroe Versatile Solid Navy Baby Tote Bag

Quantity: Total: ₹2000

Cart Summary
Subtotal: ₹3000
Shipping: ₹100
Total: ₹3100
[Proceed to Checkout](#)

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COSMOS

Admin Dashboard

- Sales analytics, product & order management

Admin Panel
Dashboard
Products
Users
Orders

Admin Dashboard

Total Products
2

Total Users
3

Total Orders
0

Admin Panel
Dashboard
Products
Users
Orders

Manage Users

Here you can view, add, edit, and delete users.

+ Add User

Name	Email	Role	Created At	Actions
abc	abc@gmail.com	user	8/20/2025	EditDelete
abc34	ab34c@gmail.com	user	Invalid Date	EditDelete
anjali15	anjali15@gmail.com	user	Invalid Date	EditDelete

9. Security & Validation

- JWT-based authentication
- Role-based access control (Admin/User)
- API validation and error handling
- Environment variables for sensitive data